

ATHARVA YEOLE

Computer Science Student · Machine Learning & Systems

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ABOUT ME

Computer Science student at Manipal Institute of Technology with hands-on experience in Python, machine learning workflows, and low-level programming in C and C++. I actively participate in competitive programming and have built several personal projects that go beyond coursework. I am drawn to problems that require careful thinking before implementation, and I strongly prefer understanding a system deeply over simply making it work.

PROJECTS & EXPERIENCE

ResNet from Scratch

Personal Project · Deep Learning / Computer Vision

- Implemented a full ResNet architecture in PyTorch from scratch, building a thorough understanding of residual connections and the difference between plain and deep residual networks.
- Conducted experiments with hyperparameters (depth, learning rate, regularisation) to adapt the model to different computer vision tasks.
- Gained practical insight into how skip connections solve the vanishing gradient problem in very deep networks.

Autograd Engine in Python

Personal Project · Systems / ML Infrastructure

- Built a scalar-valued automatic differentiation engine in Python capable of training small neural networks end-to-end, implementing forward and backward passes manually.
- Benchmarked the Python engine against a C++ port to quantify the performance gap; planning to complete and open-source the C++ version as the next milestone.
- Deepened understanding of how frameworks like PyTorch compute gradients under the hood.

EDUCATION

Manipal Institute of Technology

B.Tech in Computer Science and Engineering · Ongoing

St. Paul School (CBSE)

Grades 11 & 12

Podar International School, Besa

Schooling up to Grade 10

SKILLS

Languages: Python C C++

Libraries: PyTorch NumPy Pandas

Interests: Competitive Programming Deep Learning Systems Programming